can i get flash questions on johns lectures? my understanding is still low

Kia ora, thanks for reaching out — lots of people find John's lectures tricky at first, so you're in good company. We can absolutely do flash-style questions and build things up slowly.

John's Blood & Immune lectures cover a few chunks. Which area would you like flash questions on first?

- Innate vs adaptive immunity
- Blood components (RBCs, platelets, clotting)
- Complement, opsonisation, phagocytosis
- Antibodies and B cells
- T cells, MHC, and antigen presentation
- Hypersensitivity/allergy and monoclonal antibodies

Pick one of those (or tell me "all of them, but start easy"), and I'll start firing short questions at you, one at a time.

i think innate vs adaptable is comfortable for now

Nice, that's a great place to start. We'll keep them short and build your confidence.

I'll ask you **one question at a time**. Just give your best guess + a short reason, even if you're unsure — that's how your brain learns.

Q1

Innate immunity and adaptive immunity both protect us from infection.

Flash question 1:
Which **two key words** best describe innate immunity?

A. Specific and slow
B. Non-specific and fast
C. Specific and has memory
D. Non-specific and has memory

What do you think, and why?

Non-specific and fast

A and C describes adaptive, D describes nothing

Nice reasoning — you're spot on.

- **Innate:** non-specific + fast → **B**
- **Adaptive:** specific + has memory → **C**
- **A** is wrong because "specific" ≠ innate, and "slow" doesn't fit either.
- **D** doesn't match a real system, as you said.

Let's build a bit more.

Q2

Imagine a bacteria gets into a small cut in your skin.

Flash question 2:
Which **innate** defence would act **first**?

A. Antibodies produced by plasma cells
B. Cytotoxic T cells killing infected cells
C. Skin barrier and local phagocytes (like macrophages)
D. Memory B cells rapidly producing antibodies

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OK

Which option would you choose, and what's your short reasoning?

should be C

Yes, C is exactly what we expect first:

- **Skin barrier** is the physical innate defence.
- If the skin is broken, **local phagocytes (macrophages, neutrophils)** are among the first cells to respond.
- A, B and D all involve the **adaptive** system (antibodies, T cells, memory).

Q3

Now let's zoom in on **features** of innate vs adaptive.

Flash question 3:
Which of these is a feature of **adaptive** immunity but **not** innate immunity?

A. Present from birth
B. Responds in minutes to hours
C. Has immunological memory
D. Uses phagocytes to destroy pathogens

What do you think, and why?

should be C,
A. describes innate, B. is false. D describes both

You're doing really well with these.

You're right: **C (has immunological memory)** is the key feature of **adaptive** immunity.

Tiny tweaks:

- **A. Present from birth** → this is more true for **innate** (adaptive develops after exposure).
- **B. Responds in minutes to hours** → that's **innate**; adaptive usually takes days on first exposure.
- **D. Uses phagocytes** → mainly innate, but adaptive can "direct" them via antibodies and cytokines, so there's overlap.

Q4

Which of the following is an example of an **innate** immune component?

A. IgG antibody
B. Complement proteins
C. Memory T cells
D. A highly specific antibody response after vaccination

Pick one and tell me your reasoning.

im unsure but its A or B, im leaning A

Nice job being honest about feeling unsure — that's where the good learning happens.

Here, the best answer is **B. Complement proteins**.

Quick rundown:

- **Complement proteins** = part of the **innate** system. They're always present in blood (just need activating) and act in a **non-specific** way (e.g. opsonisation, membrane attack complex).
- **IgG antibody** = **adaptive** (made by plasma cells after specific antigen exposure).
- **Memory T cells** = adaptive.
- **Specific antibody response after vaccination** = classic adaptive response.

Next one (still on innate/adaptive):

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OK

Q5

Which cell type is **most strongly associated with innate immunity**?

A. Neutrophil
B. B cell
C. Plasma cell
D. Cytotoxic T cell

Which would you pick, and why?

A

Yes, **A. Neutrophil** is the one.

Neutrophils are classic **innate** immune cells:

- They respond **quickly** to infection.
- They're **non-specific** phagocytes (eat almost anything "non-self").
- They're often the first white cells at sites of acute inflammation.

B cells, plasma cells, and cytotoxic T cells all belong to the **adaptive** system.

Let's push your understanding a little further.

Q6

Think about **how** innate immunity recognises pathogens.

Innate immune cells like neutrophils and macrophages use receptors (e.g. TLRs) that detect common features on microbes.

Flash question 6:
Are these receptors:

A. Highly specific for one unique antigen
B. Pattern-recognition receptors that recognise common microbial patterns
C. Only present on B cells
D. Only present after vaccination

Which option fits best, and what's your reasoning?

It should be B, A and D associate with memory and specificity which is adaptive, and B cells are adaptive

Exactly, **B** is the one.

- Innate cells (like macrophages) use **pattern-recognition receptors** that recognise **common microbial patterns** (e.g. LPS on Gram-negative bacteria), not one unique antigen.
- **A & D** = very specific and often memory-based → that's adaptive.
- **C** wrong because B cells are adaptive; innate cells (macrophages, dendritic cells, etc.) also have these receptors.

You're actually reasoning really well now.

To check if the "big picture" is clicking:

Q7

If you get infected with a virus **for the second time**, and you clear it much faster than the first time, which system is mainly responsible for that faster, stronger response?

Answer in your own words (no options this time).

adaptive

Yes, that's it — the **adaptive** immune system.

On second exposure, **memory B cells and T cells** kick in quickly and strongly, which is why you clear the infection faster. That memory is the big superpower of adaptive immunity (and the whole basis of vaccination).

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You've got a solid grasp of the differences now:

- Innate: fast, non-specific, no memory
- Adaptive: slower first time, specific, has memory

Where would you like to go next with John's material?

- Still more practice on innate vs adaptive
- Blood components (RBCs, platelets, clotting)
- Complement, opsonisation, phagocytosis
- Antibodies and B cells
- T cells and MHC

Pick one, and I'll make more short flash questions for that topic.

what is most foundational? you pick from there. i am less confident on these topics

Nice question — thinking about "what's foundational" is very BIOSCI-107 of you.

For John's material, a good building order is:

1. **Blood components** (what's in blood, what WBCs exist)
2. Then **innate tools**: complement, opsonisation, phagocytosis
3. Then **antibodies & B cells**
4. Then **T cells & MHC**

So let's start with **blood components**, because everything else plugs into that.

Flash question 1 (Blood basics)

Blood has three main "formed elements".

Can you name **two** of them (just in words, not detail), and briefly say what each does?
(e.g. "X — does Y").

unsure

That's totally fine — this is exactly what we're here for. Let's build it from scratch.

The three main **formed elements** in blood are:

1. **Red blood cells (RBCs / erythrocytes)**
   - Main job: carry **oxygen** (via haemoglobin) and help carry some CO_2.
2. **White blood cells (WBCs / leukocytes)**
   - Main job: **immune defence** (innate and adaptive).
3. **Platelets (thrombocytes)**
   - Main job: help with **clotting** when a blood vessel is damaged.

Quick check:

Question:
If a person has very low platelets, what process would be most affected:
oxygen transport, immune defence, or blood clotting?
Answer with one phrase and a short reason.

im not confident here mate, can we go back to mcqs

All good, we can definitely stick with MCQs — heaps of students find that format easier to build confidence with.

We'll reset and come at the same idea in MCQ form.

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Blood basics MCQ 1

Which blood component is **most directly** involved in forming a blood clot after a vessel is damaged?

A. Red blood cells
B. White blood cells
C. Platelets
D. Plasma proteins only

Which option do you think it is, and what's your best guess why?

unsure

That's okay — let's treat this as a "guided guess" rather than a test.

We know:

- **Red blood cells** mainly carry oxygen.
- **White blood cells** are for immune defence.
- **Platelets** and some **plasma proteins** (like fibrinogen) are involved in clotting.

Given that, which seems **most directly** responsible for forming the clot structure at the injury site:

A. Red blood cells
B. White blood cells
C. Platelets
D. Plasma proteins only

Have a go at picking **one letter** now, even if it feels like guessing. We'll use your choice to anchor the explanation.

C

Nice, **C** is the one to go for.

- **Platelets** stick to the damaged vessel wall, get activated, and form a **platelet plug**.
- Then they help activate the **coagulation cascade**, which turns fibrinogen (a plasma protein) into **fibrin**, forming a stable clot.

So platelets are the **key formed element** in clotting.

Let's keep it MCQ and simple.

Blood basics MCQ 2

Which statement about **red blood cells (RBCs)** is **true**?

A. They have a nucleus so they can divide in the bloodstream
B. They mainly transport oxygen using haemoglobin
C. They are a type of white blood cell
D. They are only involved in blood clotting

Pick the option you think is most likely, and tell me your reasoning in a short sentence.

A, no nucleus and they dont divide i think they just circulate or something, so A is wrong.

B, im unsure

C, silly and false

D, I think that was platelets

Must be B

Your reasoning there was excellent.

- **B is correct:** red blood cells mainly transport **oxygen using haemoglobin**.
- **A:** as you said, RBCs **do not have a nucleus** and **do not divide** in the bloodstream.

- **C:** RBCs are not white blood cells.
- **D:** clotting is mainly **platelets + clotting factors**, not RBCs.

You're actually piecing this together really nicely.

Let's bring white blood cells into the picture.

Blood basics MCQ 3

Which of these is a **white blood cell (WBC)** involved in **innate** immunity?

A. Neutrophil
B. Red blood cell
C. Platelet
D. Plasma cell

Which option do you think fits best, and why?

A

Yes, **A. Neutrophil** is right.

- **Neutrophils** = white blood cells, part of the **innate** immune system, great at **phagocytosis** and rapid responses.
- **Red blood cell** → oxygen transport.
- **Platelet** → clotting.
- **Plasma cell** → an activated B cell that secretes antibodies (so **adaptive**, not innate).

You've now correctly identified the key formed elements and linked them to their main roles — that's a big foundational step.

Next, let's gently connect blood to immunity.

Blood + immune MCQ 4

Which pair is correctly matched?

A. Platelets — oxygen transport
B. Neutrophils — innate immune defence
C. Red blood cells — antibody production
D. Plasma cells — blood clotting

Choose one letter and tell me your short reasoning.

only b

Yes, **only B** is correctly matched:

- **Neutrophils — innate immune defence** ✅
- Platelets → clotting, not oxygen
- Red blood cells → oxygen, not antibodies
- Plasma cells → antibody secretion, not clotting

You're consistently matching functions now — that's exactly the "foundation" we wanted.

How about we move one small step deeper into immunity:

Innate vs adaptive + blood MCQ 5

Which of the following is **both** a white blood cell **and** part of the **adaptive** immune system?

A. Neutrophil
B. B cell
C. Red blood cell
D. Platelet

Pick one and tell me your reasoning.

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OK

B cell

Exactly, **B cell** is the adaptive white blood cell here.

- **B cells** = lymphocytes, part of the **adaptive** immune system, can become **plasma cells** that secrete antibodies.
- **Neutrophils** = WBC but **innate**.
- **Red blood cells** & **platelets** = not WBCs.

You've now:

- Linked RBCs → oxygen
- Platelets → clotting
- Neutrophils → innate
- B cells → adaptive

That's a really solid base.

Would you like the next MCQ set to focus on:

1. **Complement, opsonisation, phagocytosis**, or
2. **Antibodies & B cells** (going deeper into B cells)?

Pick one, and we'll keep going with short MCQs.

yes lets try complement (im scared of this one the most)

Love that you picked the scary one — complement is less evil than it looks once we chunk it.

I'll keep it simple and MCQ-based.

Big picture: complement = a set of plasma proteins that, once activated, help kill microbes by:

- Opsonisation (tagging)
- Forming pores (MAC)
- Recruiting inflammation

Complement MCQ 1

Which statement best describes the **complement system**?

A. A group of plasma proteins that help destroy microbes when activated
B. A type of white blood cell that produces antibodies
C. A hormone released by the pancreas during infection
D. A structural protein inside red blood cells

Which option would you pick, and what's your short reasoning?

A?

Yes, **A** is exactly right.

- Complement = a **group of plasma proteins** (mostly made by the liver)
- They circulate in blood in an **inactive** form and, when activated, help **destroy microbes** (innate immunity).

B, C, and D don't describe complement at all.

Now let's zoom in on one of complement's key jobs.

Complement MCQ 2

"Opsonisation" by complement means:

A. Directly forming pores in the microbe's membrane
B. Tagging a microbe to make it easier for phagocytes to eat

C. Producing antibodies that bind to pathogens
D. Activating T cells in lymph nodes

Which option seems most correct to you, and why?

is it C

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OK